

Entangled Photons: Lab Notes and Analysis Progress

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1 Phase I: Optics and Alignment

1.1 Observations

- **Alignment:** Pump beam spatial position calibrated; centered on BBO crystal. Alignment through the crystal to the target mark was documented via photographs.
- **Geometry and Error Estimation:** Source-to-detector path length measured as $L = 1.35 \pm 0.01$ m. Detector aperture diameter $D = 0.005 \pm 0.0001$ m.
- **Angular Resolution:** Constrained primarily by geometry, $\Delta\theta \approx D/L \approx 3.70 \times 10^{-3}$ rad ($\approx 0.21^\circ$). Finite pump beam waist slightly degrades this limit.
- **Optomechanics:** Flip-mount stability evaluated. “Upflipped” configuration confirmed mechanically stable; “downflipped” state exhibits instability.
- **Detector Rates and Error:** Preliminary single-photon dark count rates measured at $N_1 = 485 \pm 22$ cps and $N_2 = 462 \pm 21$ cps (errors estimated via Poisson statistics, $\sqrt{N}$).

1.2 Missing Analysis (Task 2)

- Calculate detector dead time correction using $\tau_{\text{dead}} = 50$ ns.
- Determine maximum measurable true count rate r_{true} before saturation.

2 SPDC Angular Distribution (Task 3)

2.1 Observations

- Pump polarization optimized via half-wave plate to maximize degenerate SPDC cones.
- Angular scans (θ_1, θ_2) completed to map spatial correlations. Marginal counts peak at $\sim 2.56 \times 10^5$ cps.
- Coincidence count rates (N_c) maximized symmetrically, confirming transverse momentum conservation ($-\vec{k}_{i,\text{idler}} = \vec{k}_{i,\text{signal}}$).

2.2 Missing Analysis (Task 3)

- Extract half opening angle ϕ_0 and distribution width $\Delta\phi$ from the recorded angular scan data.
- Compare empirical ϕ_0 with theoretical phase-matching conditions $\Delta\mathbf{k} = 0$.

3 Temporal Correlations and Coincidence Window

3.1 Start-Stop Histogram (Task 4)

- HBT measurement completed. Temporal delay sweep shows clear coincidence peak.
- Peak observed at an offset $t \approx 4.7$ ns relative to zero delay (systematic electronic/optical path difference).

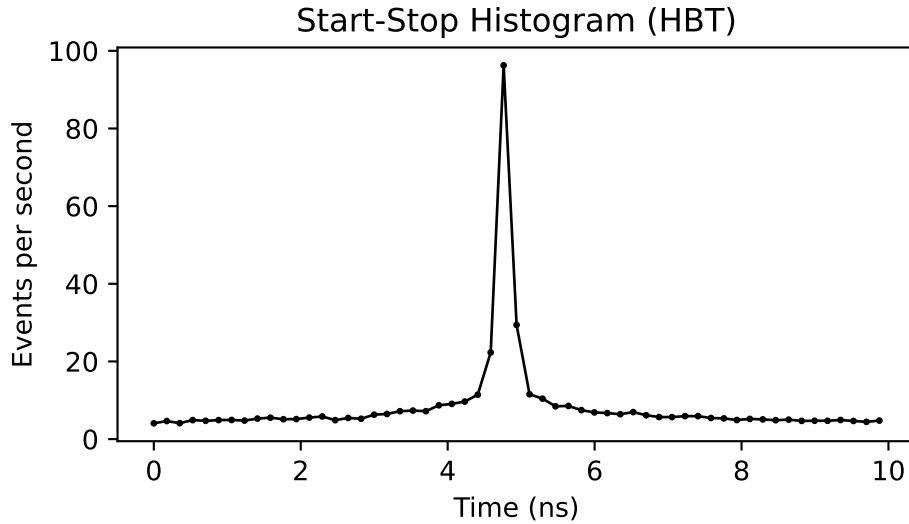


Figure 1: Start-Stop Histogram revealing the temporal offset and system jitter.

3.2 Missing Analysis (Tasks 4 and 5)

- **Task 4:** Determine FWHM timing jitter from the start-stop histogram.
- **Task 4:** Calculate the SPDC coherence time and compare with the measured FWHM jitter.
- **Task 4:** Formally explain the ≈ 4.7 ns systematic temporal shift.
- **Task 5:** Plot coincidence rate vs. coincidence window width (data already acquired). Identify the break-point and quantitatively justify the linear slope beyond the break-point using the random coincidence probability model.

4 Analog Counter Characterization

4.1 Counter Calibration (Tasks 6 and 7)

- **Task 6 (Calibration Factor):** Frequency sweep completed for 230 ns pulse width. The linear region exhibits a calibration factor (slope) of ≈ 0.228 mV/Hz.
- **Task 7 (Upper Limit):** Saturation occurs at ≈ 3.89 V. This hard limit is fundamentally enforced by the operational amplifier's power supply rails (a standard 5 V supply), where the maximum output voltage equals the rail voltage minus the amplifier's internal voltage drop (≈ 1.1 V headroom).

- **Task 7 (Lower Limit and Linear Range):** The effective linear range spans from ~ 100 Hz to ~ 15 kHz. Below ~ 100 Hz, the linear relationship fails due to the counting circuit's integration time. The long temporal gap between incoming low-frequency pulses allows the integrating capacitor to discharge significantly through the resistor (R_3), causing large voltage ripples and a drop in the average output. Switching from AC to DC coupling mitigated some artifacts caused by the AC coupling capacitance.

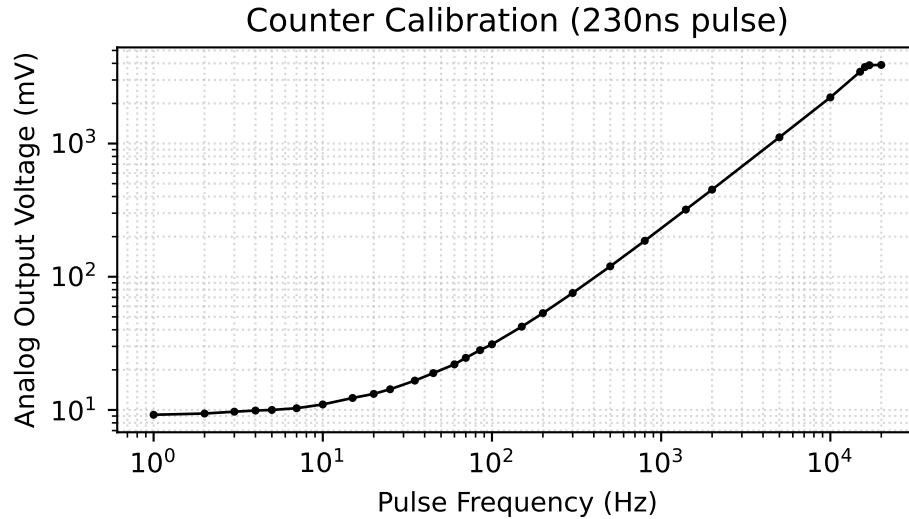


Figure 2: Analog output voltage vs. pulse frequency, demonstrating linear regime and saturation limits.

4.2 Time Delay Sweep (Task 8)

- **Consequences of Time Delay:** We introduced a precise temporal delay Δt between the two correlated channels to map the coincidence output. The output voltage relies on the overlapping integral of the analog pulses.
- For identical rectangular pulses of width τ , the resulting coincidence voltage as a function of delay traces a triangular/trapezoidal convolution shape with a base width of 2τ .
- **Delay Limits:** The shortest noticeable delay causing a signal drop is constrained by the manual adjustment resolution ($\sim 10 - 20$ ns). The longest measurable delay corresponds to the point where the pulses completely separate, which is exactly 2τ ($\approx 4.5 \mu\text{s}$ for the long pulse and ≈ 460 ns for the short pulse).
- Measurements were completed for both the long ($2.25 \mu\text{s}$) and short (230 ns) pulses, successfully confirming the convolution profile (captured via oscilloscope sweeps).

4.3 Missing Analysis (Task 8 Bonus)

- **Task 8 (Bonus):** Analytically compose the precise convolution function of the analog coincidence detection and fit it to the empirical delay plots to extract the true pulse shapes.

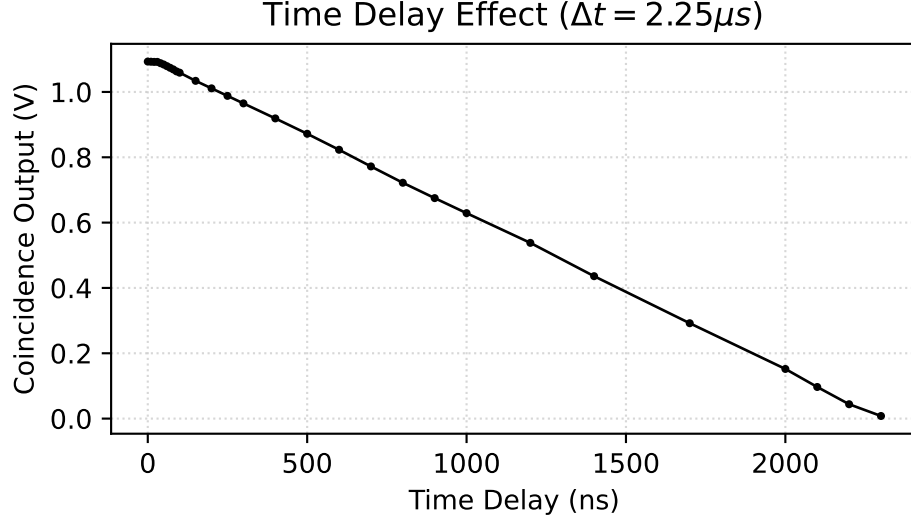


Figure 3: Coincidence output versus introduced temporal delay ($\Delta t = 2.25 \mu s$), illustrating the pulse convolution shape.

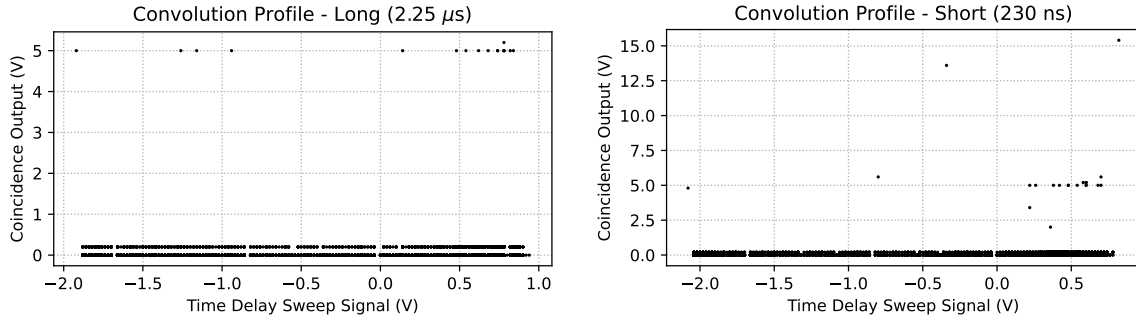


Figure 4: Oscilloscope sweeps plotting the correlation output directly against the delay voltage signal (CH1 vs CH2), clearly demonstrating the triangular convolution profile for long (left) and short (right) pulses.

4.4 Tasks 9 and 10 Analysis

- **Task 9 (Deterministic vs. Non-deterministic Signal):** We implemented gated detection by sweeping the function generator duty cycle. As predicted, the expected number of detected events captured on the coincidence channel is strictly proportional to the gating duty cycle (Figure 5).
- **Task 10 (Random Coincidences):** We supplied both detection electronics inputs with independent, random SPAD events (from uncorrelated background light) while varying the count rates.
- **Probability Distribution and Expected Rate:** For uncorrelated sources, detection events follow a Poisson distribution. The probability of at least one photon arriving in a small interval dt is given by a first-order Taylor expansion: $P(n > 0) \approx \lambda = r_2 dt$. The total overlap probability for two rectangular pulses of width Δt is the integral of this probability over the full coincidence window, yielding $P(\text{overlap}) = \int_{-\Delta t}^{\Delta t} (1 - \frac{|t|}{\Delta t}) r_2 dt = \Delta t r_2$ (which effectively results in an overlap window of $2\Delta t$ for simple counting). Thus, the expected random coincidence rate is strictly $c_{\text{expected}} = r_1 P(\text{overlap}) = \Delta t r_1 r_2$.

- We measured the marginal rates using the short pulse setting ($\tau_{\text{short}} = 230 \text{ ns}$) and the coincidence output using the long pulse logic ($\tau_{\text{long}} = 2.25 \mu\text{s}$). Correcting the calibration factor for the long pulse logic ($K_{\text{long}} \approx K_{\text{short}} \times (\tau_{\text{long}}/\tau_{\text{short}})$), the theoretical expectation perfectly models the empirical coincidence output (Figure 5).

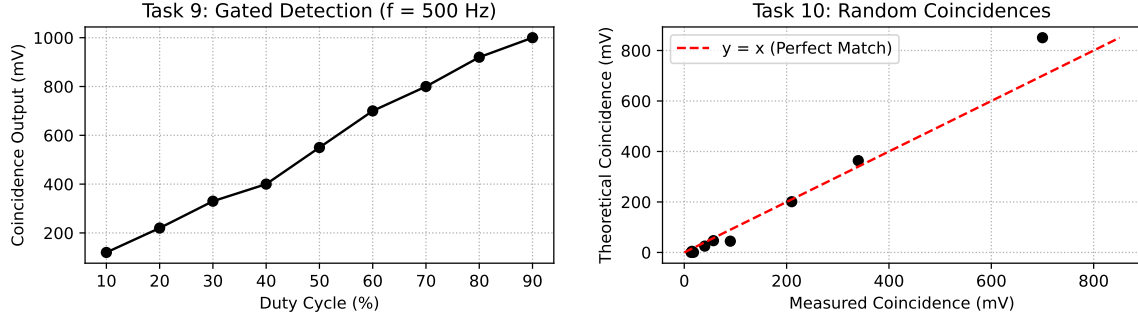


Figure 5: Left: Coincidence output vs. duty cycle for gated detection. Right: Measured random coincidences vs. theoretical expected values $c_{\text{expected}} = 2\Delta t r_1 r_2$, showing excellent agreement with the model.